

Question 8

- A database management system (DBMS) is a specialised software system that is designed to store, retrieve and manipulate data. It allows users and applications to create, read, update and delete data.

- The four popular DBMS are MySQL, PostgreSQL, Oracle and MongoDB and each is briefly explained below:-

① MySQL - it is mostly useful where there are no large amounts of data involved. For example, for small web-based solutions like building an e-commerce store platform.

② Oracle - it is a relational DBMS that supports multiple data models such as documents, graphs, relational and key-values within a single database. It is owned by Oracle Corporation. An example is a financial institution where it is used to manage sensitive financial data transactions.

③ PostgreSQL - it is an object relational DBMS where users define objects. Table approaches are combined to build more complex data structures. It can process workloads for both single machine products and complex applications. It is owned by PostgreSQL Global. An example of a user case is when building a database automation tool.

④ MongoDB - It is a free open source non relational DBMS that can process multiple data types simultaneously using internal cache. Example of a user case is developing a product catalog.